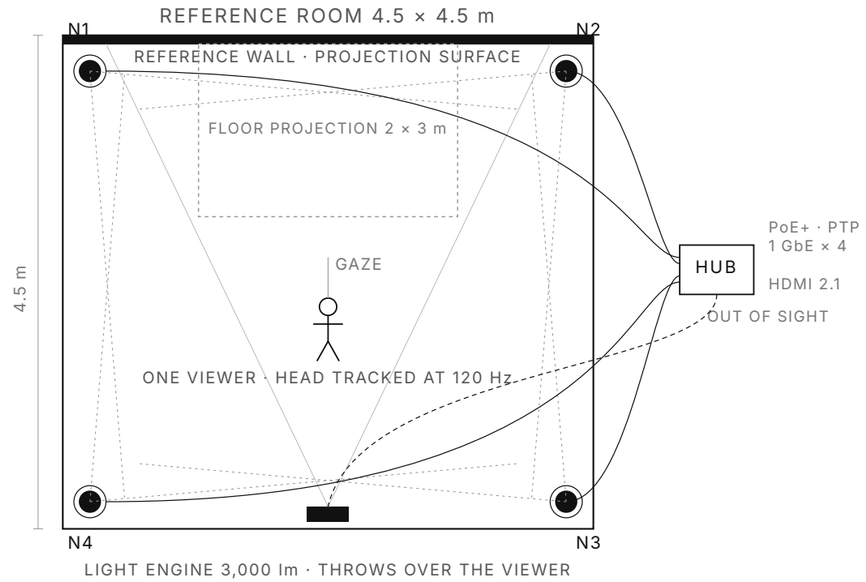
Design notes



FOUR EYES, ONE MIND.

DESIGN NOTES · MII / 02

mii is not a device you look at. It is four quiet columns at the corners of a room, a hub you never see, and light that falls on the walls you already own. The columns only sense; the hub alone thinks; the display is whatever light engine the room can carry today — and something better tomorrow.

Every part is replaceable without touching the others. The sensing core is fixed for a decade. The display layer is changed like a lens. The hub runs the same RockstarOS as a phone with an eSIM, so the room continues the conversation you started on the train.

4+1
NODES + HUB

120 Hz
SYNCHRONOUS EXPOSURE

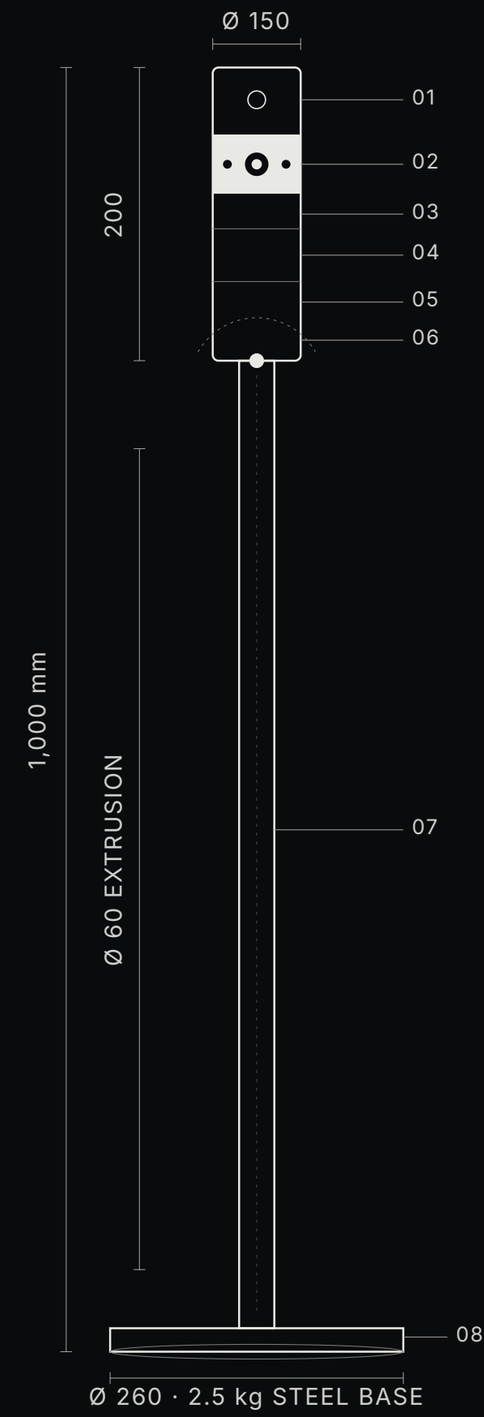
≤100 µs
INTER-NODE SKEW

3–6 m
ROOM, PER SIDE

±10 mm
JOINT ACCURACY, CENTRE

≤30 ms
MOTION TO PHOTON

Anatomy · the node



01 · CONTEXT	RGB fisheye, 160–180°, 60 fps. Room, furniture, tools, people as meaning.
02 · TRACKING	Global-shutter NIR, 1920 × 1280, 120 fps, 100° lens, 850 nm filter. Head region sent at full resolution; body at 2 × 2 binning.
03 · ILLUMINATION	850 nm pulse, 3 W radiant, fired only inside the exposure. Hardware current limit, watchdog, thermal cut-off.
04 · RADAR	60 GHz FMCW, 3Tx / 4Rx. Presence, range and velocity when the cameras lose a limb. Time-division scheduled by the hub.
05 · AUDIO · IMU	Two MEMS microphones, 6-axis IMU. Beam-formed “Hey, Eve”; tilt detection cuts IR and radar in 100 ms.
06 · SHUTTER	Mechanical lens shutter. Closed, the room is sensed by radar only.
07 · COLUMN	Cables inside the Ø 60 extrusion. PoE+ (25.5 W) or 12 V. 12 W average, no fan, surface ≤ 45 °C.
08 · BASE	Thin steel plate, Ø 260, 2.5 kg. Static tip-over resistance 4.8 N at 1 m. Floor-screw option.

WHAT A NODE DOES NOT DO

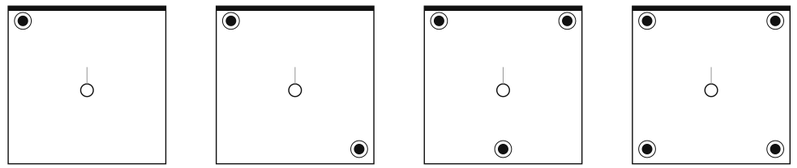
It does not think, store, or connect to anything but its hub. No battery, no Wi-Fi, no operating system. Four nodes are one instrument; the hub is the only mind in the room.

Three forms — Column, Puck (ceiling), Slot (wall) — share the same head, board and firmware. Only the mount changes.

FROM ONE TO FOUR.

DESIGN NOTES · MII / 03

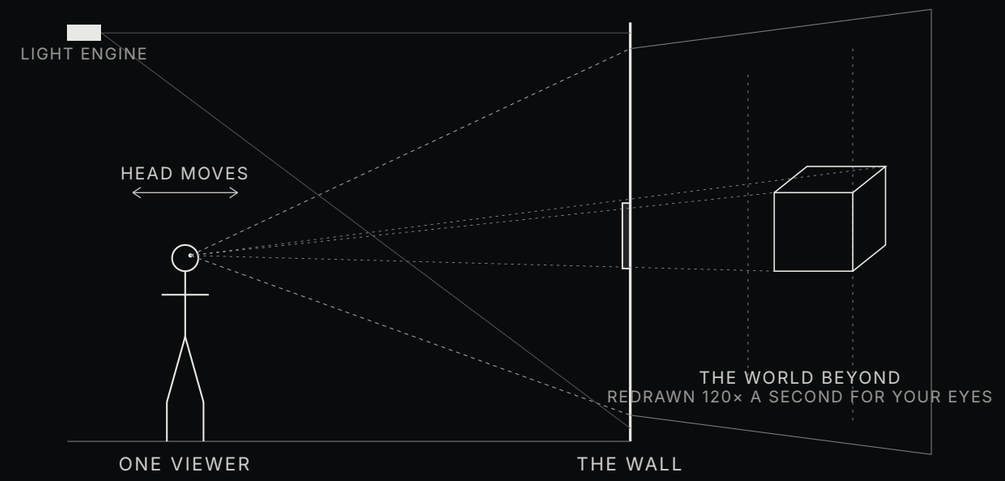
A room begins with one column and grows. Every column added is a new point of view the hub fuses immediately; nothing is replaced, nothing is reconfigured. What changes is how much of you the room can see — and therefore what it can do.



ONE	Where you are, how you move, where your head is. Eve's voice and face. Motion parallax already works — the world stays inside the wall.
TWO	Diagonal pair, longest baseline. True depth by triangulation; eyes located to ±10 mm. Objects may stand on the floor and lean 30 cm out of the wall.
THREE	Turn your back and tracking holds. The renderer keeps your own shadow off the thing you are looking at.
FOUR	The whole body, all round. Guides projected onto your own limbs; the tip of a shinai at 10 m/s; a life-size partner in another city standing where you stand.
EIGHT	Rooms beyond six metres, several people, a factory floor. Same hub family, same calibration, same API.

Every node is an equal. There is no master column — only the hub, which keeps the clocks, schedules the radar, picks the best two to four views for each joint, and tells the software one word: *presence, pose2d, pose3d, robust, full.*

Take a column away and the room degrades gracefully: the picture never stops, the word changes, Eve says so. Add one and calibration takes a minute — hold the target, walk once round the room.



No glasses. No screen. A wall that isn't a wall.

Generation one draws the world the way a window would show it. Four columns know where your eyes are to a centimetre and where they will be in thirty milliseconds. The light engine redraws the wall and the floor from exactly that point, so the room beyond the wall keeps its depth as you walk, and a tetrahedron on the floor stays put when you circle it. For one person it is immersion; for a second it is a clever painting — so remote play puts one player in each room.

Generation two adds a real aerial image — thirty to fifty centimetres of light that exists in the middle of the room, formed by retro-reflective optics in the heads of the columns and shared by everyone present. The columns do not change. Only the light does.

3,000 lm
ONE ENGINE · ONE WALL

≥150 lux
ON THE WALL, LIGHTS DIMMED

≤5 mm
GEOMETRIC ERROR AT THE CORNERS

50 cm
DEPTH IN FRONT OF THE WALL

8 /10
WHO SAY THE ROOM CONTINUES

Gen 2
AERIAL IMAGE · THE SAME COLUMNS